

PathCompanion AI

App Pages Guide — Ovvoru Page-um Enna Seiyuthu, Eppadi Use Panrathu
(English + Tamil explanation for demo & viva preparation)

CIS 6035 Development Project — An Agentic Career and Talent Companion System

1. Login / Sign-up

WHAT IT DOES (English)

The entry point of the application. Users register with an e-mail and password through Supabase Authentication. The moment a new user signs up, a database trigger automatically creates their profile row. Row-Level Security guarantees each user can only ever see their own data.

ENNA SEIYUTHU (Tamil)

Ithu app-oda entry point. Email + password vachchu register pannalaam — Supabase Authentication ithai handle pannuthu. Pudhu user sign-up panna odane, database trigger automatic-ah avangalukku profile row create pannidum. Row-Level Security irukkurathala oru user innoru user-oda data-vai paakkave mudiyaathu.

HOW TO USE (English)

- Open the app — the Login page appears first.
- Click 'Sign up', enter e-mail and password, and submit.
- Check the e-mail for confirmation if required, then log in.
- You are redirected to the Dashboard; a profile has been auto-created.

EPPADI USE PANRATHU (Tamil)

App-ah open pannaa Login page than mothalla varum. 'Sign up' click panni email/password kudunga. Login aana odane Dashboard-ku pogum — profile automatic-ah create aagirukkum.

DEMO TIP: Show sign-up briefly, then use an existing account with data already added — the assessment brief says proper data must be prepared beforehand.

2. Dashboard

WHAT IT DOES (English)

The home screen after login. It shows system health (Backend + Database: connected) and quick statistics gathered from all agents — number of job matches, open skill-gap analyses, learning progress and recent activity. It is the one-glance summary of the user's career journey.

ENNA SEIYUTHU (Tamil)

Login aana odane varura home screen. System status (Backend + Database connected) plus ella agents-oda quick stats kaatuthu — evlo job matches, evlo open skill gaps, learning progress, recent activity. Un career journey-oda one-glance summary ithu.

HOW TO USE (English)

- Log in — the Dashboard loads automatically.
- Read the status cards: backend connection, database connection.
- Review quick stats; click any card to jump to that agent's page.

EPPADI USE PANRATHU (Tamil)

Login pannina odane Dashboard automatic-ah load aagum. Status cards-ah paarunga — backend/database connected-ah nu. Entha card-ah click pannaalum antha agent page-ku direct-ah pogalaam.

DEMO TIP: Point at 'Backend + Database: connected' — it proves the full stack is live before you start the demo.

3. Profile & Skills

WHAT IT DOES (English)

Here the user builds their master profile: target role, location, and skills added as chips, each with a proficiency level. This profile is the system's shared memory — every agent reads from it. Each skill also records its source: added manually, extracted from a resume, or earned through the Learning agent.

ENNA SEIYUTHU (Tamil)

Inga user master profile build pannuvaanga: target role, location, skills-ah chips-ah add pannalaam, ovvonnukkum proficiency level irukku. Ithu thaana system-oda shared memory — ella agents-um ithai read pannum. Ovvoru skill-kum source track aagum: manual-ah add pannatha, resume-la irunthu vanthatha, illa Learning agent moolama kaththatha.

HOW TO USE (English)

- Open Profile & Skills from the sidebar.
- Set your target role (e.g., 'Frontend Developer') and location.
- Type a skill name and add it — it appears as a chip; set proficiency.
- Save; every other agent now uses this profile.

EPPADI USE PANRATHU (Tamil)

Sidebar-la Profile & Skills open pannunga. Target role (example: 'Frontend Developer') and location set pannunga. Skill name type panni add pannaa chip-ah varum. Save pannina odane matha ella agents-um intha profile-ah use pannum.

DEMO TIP: Add 8–10 real skills beforehand — richer profile means better match scores in Job Scout, which makes the demo look strong.

4. Job Scout

WHAT IT DOES (English)

Aggregates real job postings from public APIs (e.g., Remote). 'Sync Jobs' fetches fresh postings and stores each with a 768-dimension semantic embedding. 'Find my matches' runs hybrid matching — embedding similarity plus skill overlap — and assigns every job a match percentage with matched and missing skill tags. 'Analyze gap' on any job card sends that job straight to the Skill Gap agent.

ENNA SEIYUTHU (Tamil)

Public APIs-la irunthu (Remote maathiri) real job postings-ah kondu varuthu. 'Sync Jobs' pudhu jobs-ah fetch panni 768-dimension embedding-oda store pannum. 'Find my matches' hybrid matching odum — meaning similarity + skill overlap — ovvoru job-kum match percentage, matched/missing skills tags kudukkum. Job card-la 'Analyze gap' click pannaa antha job direct-ah Skill Gap agent-ku pogum.

HOW TO USE (English)

- Click 'Sync Jobs' to pull fresh postings (needs a moment; runs in background).
- Click 'Find my matches' — jobs are ranked by match percentage.
- Use filters (Remote only / Scored matches) and the search box.
- Open 'View Details' for the full posting, or 'Analyze gap' to jump to Skill Gap.

EPPADI USE PANRATHU (Tamil)

'Sync Jobs' click pannaa pudhu jobs varum (background-la odum, UI block aagaathu). 'Find my matches' click pannaa match percentage order-la jobs varum. Filters/search use pannalaam. 'Analyze gap' click pannaa antha job-oda Skill Gap analysis-ku direct-ah pogalaam.

DEMO TIP: Explain the 66% match: 'This is semantic matching, not keyword matching — my React profile matches a Frontend Developer vacancy.' Then click Analyze gap to show agents cooperating.

5. Skill Gap

WHAT IT DOES (English)

Paste any job description (or arrive from a job card) and the custom-trained Model 2 extracts the required skills from the text. The system compares them with the user's profile and splits them into 'present' and 'missing'. Because extraction is done by the project's own trained model — not an external AI service — results are deterministic and reproducible.

ENNA SEIYUTHU (Tamil)

Entha job description-aiyum paste pannalaam (illa job card-la irunthu varalaam). Custom-trained Model 2 antha text-la irunthu required skills-ah extract pannum. User profile-oda compare panni 'present' / 'missing' nu pirikkum. Intha extraction namma own trained model thaana pannuthu — external AI illa — athanala results consistent-ah, reproducible-ah irukkum.

HOW TO USE (English)

- Open Skill Gap; paste a job description into the text box.
- Click 'Extract' / 'Analyze' — Model 2 lists the required skills.
- Review the split: skills you have vs skills you are missing.
- Click the action to generate a learning plan for missing skills.

EPPADI USE PANRATHU (Tamil)

Skill Gap page-la job description paste pannunga. Analyze click pannaa Model 2 required skills-ah kaatum. Present/missing split-ah paarunga. Missing skills-ku learning plan generate panna button irukku.

DEMO TIP: Say clearly: 'This runs on MY trained model — a multi-label DistilBERT — served locally from the backend.' This is Model 2 working live; examiners give marks for it.

6. Resume

WHAT IT DOES (English)

Paste resume text or upload a PDF. The custom-trained Model 1 — a DistilBERT classifier trained on 2,484 resumes across 24 occupational categories — instantly predicts the resume's category with top-3 confidence scores. If the predicted category does not match the target role, the system shows an alignment warning.

ENNA SEIYUTHU (Tamil)

Resume text-ah paste pannalaam illa PDF upload pannalaam. Custom-trained Model 1 — 2,484 resumes / 24 categories-la train pannina DistilBERT classifier — resume entha category nu top-3 confidence scores-oda udane predict pannum. Predicted category target role-oda match aagalannaa alignment warning kaatum.

HOW TO USE (English)

- Open Resume; paste resume text or upload a PDF file.
- Click 'Classify' — the predicted category appears with top-3 confidences.
- Check the alignment hint against your target role.
- Proceed to Resume Builder for a tailored version.

EPPADI USE PANRATHU (Tamil)

Resume page-la text paste pannunga illa PDF upload pannunga. Classify click pannaa predicted category top-3 confidence-oda varum. Target role-oda match aagutha nu paarunga. Appuram Resume Builder-ku poi tailored version generate pannalaam.

DEMO TIP: This is Model 1 live — say the numbers: '82.3% accuracy vs a 65.5% baseline on the held-out test set.'

7. Resume Builder

WHAT IT DOES (English)

Builds a job-tailored resume. It takes the master profile, aligns the content with the target job's keywords for ATS (Applicant Tracking System) optimisation, computes an ATS score, and exports the final resume as a PDF. Every job gets its own tailored version.

ENNA SEIYUTHU (Tamil)

Job-ku thagundha maathiri tailored resume build pannuthu. Master profile-ah eduthu, target job-oda keywords-oda align panni ATS optimisation pannum, ATS score compute pannum, final resume-ah PDF-ah export pannum. Ovvoru job-kum thani tailored version.

HOW TO USE (English)

- Open Resume Builder; select a target job (or paste one).
- Generate — the tailored resume draft appears with an ATS score.
- Edit any section if needed and regenerate.
- Export as PDF.

EPPADI USE PANRATHU (Tamil)

Resume Builder-la target job select pannunga. Generate pannaa ATS score-oda tailored draft varum. Venum-naa edit panni regenerate pannalaam. Last-la PDF export.

DEMO TIP: Contrast with real life: 'Most applicants send one resume everywhere; ATS systems reject them. Here every job gets its own optimised version.'

8. Learning

WHAT IT DOES (English)

For every missing skill this agent generates an ordered study plan with at least three free resources (official docs, freeCodeCamp, YouTube) and tracks progress. Key feature: marking a skill as learned fires a 'skill_learned' event — within 30 seconds the skill gap refreshes and affected resumes are flagged for refresh, automatically, without blocking the UI. This is the interconnected cascade — the project's USP.

ENNA SEIYUTHU (Tamil)

Ovvoru missing skill-kum minimum 3 free resources vachchu (official docs, freeCodeCamp, YouTube) ordered study plan generate pannuthu, progress track pannuthu. Main feature: oru skill-ah 'learned' nu mark pannaa 'skill_learned' event fire aagum — 30 seconds-kulla skill gap refresh aagum, affected resumes flag aagum — automatic-ah, UI block aagaama. Ithu than interconnected cascade — project-oda USP.

HOW TO USE (English)

- Open Learning; pick a missing skill (or arrive from Skill Gap).
- Generate plan — an ordered list of free resources appears.
- Tick resources as you complete them; progress bar updates.
- Mark the skill as learned — watch the cascade update Skill Gap and Resume.

EPPADI USE PANRATHU (Tamil)

Learning page-la missing skill select pannunga. Plan generate pannaa free resources ordered-ah varum. Complete panna panna tick pannunga. Skill-ah 'learned' nu mark pannaa — cascade Skill Gap-aiyum Resume-aiyum automatic-ah update pannurathai paarunga.

DEMO TIP: THE key demo moment: mark a skill learned, then open Skill Gap (refreshed) and Resume (flagged). Say: 'agents never call each other — they communicate through an event bus in shared memory.'

9. Career Path

WHAT IT DOES (English)

Maps long-term career progression — e.g., Intern → Junior → Mid → Senior → Lead. For each stage it shows required skills, an estimated duration in months, and practical tips. It answers the question most tools ignore: what comes next in my career?

ENNA SEIYUTHU (Tamil)

Long-term career progression-ah map pannuthu — example: Intern → Junior → Mid → Senior → Lead. Ovvoru stage-kum required skills, estimated duration (months), practical tips kaatuthu. Matha tools ignore pannura kelvi — 'en career-la aduthu enna?' — athukku answer kudukkuthu.

HOW TO USE (English)

- Open Career Path; enter your goal role.
- Generate — the stage-by-stage roadmap appears.
- Expand a stage to see its required skills and duration.
- Missing skills can flow into the Learning agent.

EPPADI USE PANRATHU (Tamil)

Career Path-la goal role enter pannunga. Generate pannaa stage-by-stage roadmap varum. Ovvoru stage-ayum expand panni skills/duration paarunga. Missing skills-ah Learning agent-ku anuppalaam.

10. Interview (Voice Mock Interview + Panic Control)

WHAT IT DOES (English)

A realistic spoken interview. The AI asks questions out loud (TTS), tailored to the actual job and the user's actual skills read from shared memory. The user answers by voice; Groq Whisper transcribes; coaching feedback follows each answer, targeting under-2-second turn latency. A panic button pauses the session and guides breathing exercises; anxiety is recorded before and after, and a full feedback report ends the session.

ENNA SEIYUTHU (Tamil)

Realistic spoken interview. AI kelvigal-ah voice-la kekkum (TTS) — shared memory-la irunthu actual job + actual skills paathu questions varum. User voice-la answer pannuvaanga; Groq Whisper transcribe pannum; ovvoru answer-kum coaching feedback varum — 2 seconds-kulla. Panic button session-ah pause panni breathing exercises guide pannum; anxiety before/after record aagum; last-la full feedback report.

HOW TO USE (English)

- Open Interview; select the job/role and set your pre-session anxiety level.
- Start — the AI speaks the first question; allow microphone access.
- Answer by speaking; view the live transcript and per-answer coaching.
- Try the panic button to see the calm-down flow.
- End the interview — review the final feedback report and anxiety change.

EPPADI USE PANRATHU (Tamil)

Interview page-la job/role select panni anxiety level kudunga. Start pannaa AI question-ah voice-la kekkum (mic allow pannanum). Pesi answer pannunga — transcript-um coaching-um varum. Panic button-ah try panni calm-down flow-ah paarunga. End pannaa final feedback report varum.

DEMO TIP: Test the microphone beforehand! Keep one question + panic button as the demo; a full interview eats your 15 minutes.

11. Mock Exam

WHAT IT DOES (English)

An extra feature beyond the original proposal. The user picks a subject; the system generates exam questions, scores the submitted answers, stores the results, and e-mails a certificate on completion.

ENNA SEIYUTHU (Tamil)

Original proposal-ku appuram add pannina extra feature. Subject select pannaa system exam questions generate pannum, answers-ah score pannum, results store pannum, mudichchathum certificate email-la anuppum.

HOW TO USE (English)

- Open Mock Exam; choose a subject and start.
- Answer the generated questions and submit.
- View the score; results are stored per user.
- Request the certificate — it arrives by e-mail.

EPPADI USE PANRATHU (Tamil)

Mock Exam-la subject select panni start pannunga. Questions answer panni submit pannunga. Score varum, results store aagum. Certificate request pannaa email-la varum.

DEMO TIP: Mention this is scope ADDED beyond the proposal — extra effort is worth saying out loud.

12. Chatbot / AI Assistant

WHAT IT DOES (English)

A conversational assistant available on every page (chat widget), powered by Groq Llama 3.3 70B for low-latency replies. It knows the user's profile context, so career questions can be asked from anywhere in the app.

ENNA SEIYUTHU (Tamil)

Ella pages-layum irukkura conversational assistant (chat widget) — Groq Llama 3.3 70B vachchu fast-ah reply pannum. User profile context athukku theriyum, so app-la enga irunthaalum career questions kekkalaam.

HOW TO USE (English)

- Click the chat bubble at the bottom-right on any page.
- Type a career question (e.g., 'What should I learn for a DevOps role?').
- The reply streams in; continue the conversation with context retained.

EPPADI USE PANRATHU (Tamil)

Entha page-layum bottom-right chat bubble-ah click pannunga. Career question type pannunga. Reply stream aagi varum; context-oda conversation continue pannalaam.